Ziang (Jay) Zhang

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Interactive AI Portfolio – Ask Me Anything: jayziangzhang.github.io/askme/

Applied AI & Data Scientist with 4+ years of experience developing GenAI applications, data platforms, and analytics solutions across financial services, manufacturing, and SaaS environments. Experienced in Python, SQL, LLMs, Retrieval-Augmented Generation (RAG), NLP applications, and large-scale data pipelines, with a proven track record of translating complex data into production-ready AI systems and measurable business impact.

SKILLS

Programming: Python, SQL, JavaScript, TypeScript
Machine Learning: Machine Learning, Predictive Modeling, NLP
Generative AI: LLMs, RAG, Embeddings, Prompt Engineering, Azure OpenAI
ML & Data Libraries: Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy
Data Science: Statistical Analysis, Feature Engineering, Data Mining, Data Visualization, Experiment Design
Sofware Engineering: REST APIs, Data Pipelines, ETL, Redis, RabbitMQ, Docker, Git, Agile/Scrum

PROFESSIONAL EXPERIENCE

Founder & Full-Stack Developer – IDP Plus | www.idp-plus.com **2025**

- Built and deployed an AI-powered SaaS platform for international driver's license translation, integrating OCR, automated document validation, and intelligent document generation into an end-to-end workflow that processed 500+ customer orders while reducing manual processing time by 95%.
- Designed scalable backend services and data-processing pipelines that integrate OCR, REST APIs, payment systems, and automated PDF generation for high-throughput document processing.
- Collaborated with external partners on printing, distribution, and marketing while independently managing deployment, website maintenance, and platform operations.

GenAI Engineer Intern, *Manulife* | *Toronto, Canada* ***2024***

- Developed a Retrieval-Augmented Generation (RAG) HR assistant POC leveraging hybrid retrieval (BM25 + embedding-based semantic search) and LLMs to enable accurate natural language access to enterprise policy knowledge.
- Designed and deployed an AI-powered HR policy authoring platform using LLMs to automate document generation, reducing drafting time by 80% while maintaining enterprise writing standards.
- Collaborated with cross-functional stakeholders to identify business problems, evaluate AI solutions, and present technical findings to business partners.

Technical Consultant – Factory Automation & Digitalization, *Siemens* | *Beijing, China* ***2021–2023***

- Delivered data solutions across 30+ manufacturing facilities — gathered requirements, scoped analytics use cases, and replaced manual reporting with automated, data-driven workflows; trained clients to self-serve monitoring and reporting.
- High-frequency Data Acquisition: Designed a high-frequency data pipeline (REST API → RabbitMQ → Redis) ingesting equipment test data every 100ms across 70+ fields; transformed raw streams into stable end-of-line test results stored in SQL for consumption by upstream systems (MES) and front-end reporting dashboards.
- Energy Platform: consolidated fragmented multi-vendor PLC and utility-meter data into an automated pipeline by designing SQL data models and database schemas, enabling energy KPIs (e.g., electricity per unit produced), interactive dashboards, and anomaly detection that supported ESG and cost-reduction initiatives.
- Software Development: Built data-quality middleware for 100+ RFID data sources, implementing deduplication, validation, heartbeat monitoring, connection state management, and message metadata processing to ensure reliable device communication and accurate inventory data in MES.

EDUCATION

University of Toronto
Master of Management Analytics, 2024
BASc, Electrical Engineering, 2021 | Minor in Engineering Business | Certificate in Artificial Intelligence